Yichao Jin, Ph.D.

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Computational Social Science / Behavioral AI / Structural Causal Modeling / Causal Inference

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Education

University of Texas at Dallas May 2026
Ph.D. in Public Policy and Political Economy

- Dissertation: "Intertemporal Choice in Vaccination Timing: Evidence from a Discrete Choice Experiment in Wuhan" — **ProQuest**
- Committee: Dohyeong Kim (Chair), Richard Scotch, Soyoung Kwon, Khai Chiong.
- Dean of Graduate Education Dissertation Research Award (2025).

University of Texas at Dallas May 2026
M.S. in Social Data Analytics and Research

University of Queensland 2019
Master of Development Economics

University of California, Riverside 2017
B.A. in Economics

Research Agenda

My research develops causally-constrained AI agents for behavioral simulation and policy analysis. I integrate discrete choice experiments, structural causal modeling, and large language models to build Behavioral Digital Twins—empirically-grounded synthetic agents that generate behavior within estimated human choice frontiers rather than from LLM training priors. This framework enables pre-screening of behavioral interventions and policy designs before field deployment, with applications to health behavior, education, and social policy.

Working Papers

1. **"Empirical-Frontier Regularization for LLM Synthetic Agents: A Preference-Anchored Framework"**
   Working Paper (Target: J. Biomedical Informatics) — **SSRN Link**. Introduces a DCE-grounded empirical-frontier regularization framework for LLM synthetic agents. Using a convex anchoring rule, the Static-BDT Anchor reduces waiting-time monotonicity violations from 62.5% to 4.2% and improves Spearman rank correlation from -0.024 to 0.952. Presented at DAISY 2026 (ACM CHIL 2026) and AIxPH 2026.

2. **"Value Contamination in Policy Simulation: Measuring How LLM Policy-Value Orientations Shape Predicted Administrative Burden Effects"**
   Working Paper — **SSRN Link**. Audit framework measuring value sensitivity in LLM-generated policy simulations. Tests nine LLMs across six model families through policy-value orientation batteries, narrative frame-decomposition, and within-model frame-manipulation experiments (N=2,160 observations).

3. **"Waiting Time as a Behavioral Barrier to Vaccination Uptake: Nonlinear Heterogeneity by Institutional Trust"**
   Under Review at Economic Analysis and Policy — **Link to Full Draft**. Structural DCE framework measuring interaction between temporal barriers and institutional trust in vaccination decisions.

4. **"The Price of Waiting: Evidence on Cash–Time Trade-Offs in Vaccination Time Discount"**
   Manuscript in Preparation (Expected submission: August 2026).

Research Experience & Affiliations

Doctoral Researcher, Spatial Health AI Research Partnership (SHARP), UT Dallas

- Leading interdisciplinary research on the intersection of artificial intelligence, geospatial analytics, and health policy.
- Managing large-scale datasets and developing computational simulations (R, Python) to evaluate pandemic preparedness.
- Leading development of the Behavioral Digital Twin (BDT) framework, integrating DCE data with causally-constrained LLM agents for synthetic policy simulation.

Graduate Researcher, Behavioral Health Economics Laboratory, UT Dallas

- Conducted high-salience behavioral experiments (N=1,027) analyzing risk-weighted health decisions and institutional trust.

Conference Presentations

- **AIxPH 2026 (1st Annual Conference on AI and Public Health, Baltimore, MD)**: Empirical-Frontier Regularization for LLM Synthetic Agents — Poster Presentation.
- **DAISY 2026 Workshop at ACM Conference on Health, Inference, and Learning (CHIL) 2026**: Behavioral Digital Twins — A Causally-Faithful Generative Framework.
- **APPAM Fall Research Conference 2025** (Seattle, WA): Estimating Time Discount Rate for Vaccination.
- **UT Dallas Graduate Research Symposium 2025**: Modeling Vaccination Decisions with Hyperbolic Discounting.

Technical Skills & Certifications

Econometrics: Mixed Logit (MXL), Latent Class Models (LCM), SEIR Modeling, Causal Inference.
AI & ML: PyTorch, Transformers (HuggingFace), LLM APIs (OpenAI, Anthropic), LangChain, Causal Machine Learning.
Software: Stata (Advanced), R, Python, MATLAB, LaTeX, SQL, GIS.
Certifications: Stanford Artificial Intelligence Graduate Certificate; CITI Human Subjects Protection.

Honors & Awards

- **2025** Dean of Graduate Education Dissertation Research Award, UT Dallas
- **2025** Betty & Gifford Johnson Graduate Travel Award, UT Dallas
- **2016** Omicron Delta Epsilon, International Honor Society for Economics

REFERENCES

Dohyeong Kim (Chair)
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Robert E. Holmes Jr. Professor of Public Policy, GIS & Social Data Analytics
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